Response to Reviewers

Manuscript: *A Dual-Track Adaptive Projection Algorithm for a Nonlinear Split Feasibility Problem with Mixed Convex–Pseudoconvex Constraints*

Journal: *Optimization Methods and Software*

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We thank the Editor and both Reviewers for their careful and constructive reading. The comments have materially strengthened the paper, and in several places led us to remove overstated claims and to add new, more honest evidence. Below we respond to each point individually and indicate the corresponding change in the revised manuscript. **All added or modified text in the revised manuscript is typeset in colored text** so the changes are easy to locate. Section, equation, theorem, and table numbers below refer to the revised manuscript. In summary, the revision:

- merges the former Introduction and Related work into a single, shorter Introduction and cuts the paper from 39 to 25 pages, rewriting it throughout in shorter sentences;
- qualifies the "constant-free" claim (new Remark 2) and reframes the proportional weight as a heuristic residual-share allocation, with no claim of necessity or optimality (new Remark 1);
- makes the whole-sequence convergence argument explicit (rewritten Stage 3 of Theorem 3.12), clarifying that it needs only one feasible cluster point and Fejér monotonicity, not a unique limit;
- adds a solver diagnostic (new Sec. 4.4) reporting per-trajectory constraint activation, a tuned-modulus subgradient comparison with its tuning cost, and controlled instances with a genuinely nonconvex pseudoconvex constraint;
- proves that the feature-selection ratio is in fact convex (new Proposition 4.1), states plainly that the application is a convex special case, and removes all claims of uniform superiority;
- unifies the reference format, adds the requested self-adaptive and decomposition works for the SFP, moves the appendix results into the main text, and resolves every notational and presentation issue raised.

Response to the Editor

The Associate Editor recorded no specific comments; the decision was major revision.

Response. We have revised the manuscript in full accordance with both Reviewers' reports, as detailed below, and have highlighted all changes in colored text. The two most substantial structural changes — merging and shortening the front matter (Reviewer 2) and adding the solver diagnostic with constraint-activation and tuned-baseline evidence (Reviewer 1) — are summarized in the list above.

Response to Reviewer #1

We thank the Reviewer for the accurate summary of the contribution and for five comments that led directly to a more careful statement of the theory and a more complete experimental section.

R1.1 The soft weight $\mu_k = J_k^{\mathrm{cvx}}/(J_k^{\mathrm{cvx}} + J_k^{\mathrm{ps}})$ is central to the algorithm's novelty, yet the convergence proof only requires μ_k to satisfy certain summability conditions – any sequence in $[0, 1]$ would suffice. The chosen form is therefore not structurally necessary.

The authors should either justify this specific weighting or reframe the contribution accordingly.

Response. We have adopted the Reviewer's second suggestion and reframed the contribution. We agree that the proportional form is not structurally necessary: the one-step quasi-Fejér inequality (Lemma 3.8) holds for every $\mu_k \in [0,1]$, and the convergence proof needs only that the weight keep each track active — with the constant weight $\mu_k \equiv \frac{1}{2}$, for instance, the proofs of Lemma 3.10 and of Stage 2 go through with only minor changes. The experiments point the same way: a balanced variant, which assigns equal weights when both tracks are violated, succeeds in 26/36 runs on the feature instances and 24/36 on the nonconvex family, the same counts as DTAP [Sec. 4.4, Table 7]. We therefore no longer present the specific weight as a theoretical necessity or an optimum. The abstract now calls it "a residual-responsive choice, without a claim of optimality", and the Introduction states that "the proof uses this weight only to keep a persistently violated track active". New Remark 1 [Remark 1, Sec. 3.4] states this explicitly: it records where the proportional form enters the present proof and notes that the one-step inequality, although valid for every $\mu_k \in [0,1]$, does not by itself imply convergence for every weight sequence. The degenerate regimes of Corollaries 3.13–3.14 correspond to $\mu_k \equiv 1$ and $\mu_k \equiv 0$. The dual-track construction — a residual-scaled step where a quantitative gradient inequality is available and a safeguarded step where only sign information is — remains the contribution; the particular weight that blends them does not.

R1.2 The "constant-free" claim is overstated. The algorithm does not explicitly require Lipschitz constants, but Assumption 1(A2) and Lemma 4.9 ensure the iterates lie in a compact set where a Lipschitz constant necessarily exists. The claim should be qualified.

Response. We agree. Such an implicit dependence is shared by any method with bounded iterates and continuous gradients, which is why we now frame the point as being about algorithm *inputs* rather than about the existence of a constant. The precise statement is now that no operator norm, Lipschitz constant, or Hölder modulus is an *input* of the iteration, while the *analysis* does use finite bounds on function values and gradients over the compact set that contains the (provably bounded) iterates. New Remark 2 [Remark 2, Sec. 3.4] makes this explicit: "Bounded gradients imply Lipschitz continuity of the functions on that convex compact set, but these bounds are not algorithm inputs," and notes that boundedness is established first (Lemma 3.9, the former Lemma 4.9) before any such bound is invoked. The abstract now says that the step sizes "require no prescribed operator norm, Lipschitz constant, or Hölder modulus" and that "the proof uses local bounds on the bounded iterate region", and the Conclusion that "no regularity modulus is an input, although the analysis uses bounds on a compact set containing the iterates". We no longer use the unqualified phrase "constant-free."

R1.3 Theorem 4.12 claims convergence of the whole sequence to a feasible point. However, Stage 2 of the proof only establishes that at least one limit point is feasible. Since the problem is nonconvex, multiple limit points may exist. The authors should either strengthen the argument or revise the theorem to subsequential convergence.

Response. We have kept the whole-sequence statement, because it is correct, and rewritten Stage 3 so that the step the Reviewer flagged is now explicit. The argument does not require a unique limit point. For *every* fixed $x^\star \in \Omega$, Lemma 3.9(i) shows that $\{\|x^k - x^\star\|^2\}$ converges (Fejér monotonicity with respect to the whole feasible set, using only $R_k^{x^\star} \geq 0$ from Lemma 3.5). Stage 2 produces *one* feasible cluster point $\bar{x} \in C \cap \Omega$. Applying Lemma 3.9(i) with the admissible choice $x^\star = \bar{x}$, the limit $\lim_k \|x^k - \bar{x}\|^2$ exists; since a subsequence converges to $\bar{x}$, that limit is 0, so the *whole*

sequence converges to $\bar{x}$. Multiple feasible cluster points therefore cannot coexist. The rewritten Stage 3 states this in full and closes with: "The argument does not require Ω to be a singleton: one feasible cluster point and the convergence of its distance suffice" [proof of Theorem 3.12, Sec. 3.8]. We have also aligned the Introduction (contribution C2) with this wording.

R1.4 The thresholds b_i are set as percentiles of sampled $F_i(w)$ to guarantee $\Omega \neq \emptyset$. In real applications, thresholds are given by the problem, not adjustable. More critically, if b_3 is so large that $F_3(w) \ll b_3$ throughout, the pseudoconvex constraint is never binding. The authors should report constraint activation frequencies and confirm the mixed setting is genuinely engaged.

Response. We now address both halves of this comment directly, in the new subsection *Constraint activity and a tuned subgradient comparison* [Sec. 4.4]. On thresholds: we clarify that externally specified thresholds remain fixed inputs, and that the percentile calibration is used only to define the benchmark instance, with the same thresholds shared by all solvers; consistency ($\Omega \neq \emptyset$) is treated as an assumption, and the exact levels are recorded in the supplementary metadata [Sec. 4.1.3 and Sec. 4.4]. On activation: we report the frequency of $F_3(w^k) - b_3 > 10^{-6}$ along every trajectory. The third constraint is in fact *inactive* on seven of the eight datasets and active at only 1 of 48 evaluated points on Abalone (Table 6). We state the consequence plainly, agreeing with the Reviewer: on these instances the pseudoconvex track is essentially never binding, so the dual-track mechanism is not needed there — these feature-selection instances exercise a convex subclass. Indeed, new Proposition 4.1 proves that the fractional criterion $F_3 = w^\top Q_R w / \rho^\top w$ is *convex* on a neighbourhood of the simplex when $Q_R \succeq 0$ and $\min_k \rho_k > 0$, so the feature model is an all-convex normal-box NSFP. To engage the mixed setting genuinely, we added controlled instances with a provably nonconvex pseudoconvex constraint, $F_3(x) = \kappa \log(1 + \|x - e_1\|^2)$ [Eq. (47), Sec. 4.4]; there the pseudoconvex track is active in all 36 runs (pooled activity 12 835/12 877 evaluated points), so the pseudoconvex track is exercised directly rather than vacuously.

R1.5 DTAP and SG-const/SG-dyn are statistically indistinguishable on three of four metrics (Table 5). SG-dyn's failure on 3 datasets is attributed to poor Hölder modulus estimation — but with a tuned modulus it might perform comparably. An additional experiment with manually tuned moduli would strengthen the comparison.

Response. We added exactly this experiment [Sec. 4.4, Tables 6–8]. For SG-dyn we now report, alongside the sampled-modulus variant, a *tuned* variant whose modulus is selected from a manual grid $c \in \{10^{-3}, \ldots, 10^{1}\}$ on held-out starts, and we account for its tuning cost separately (179 744 optimization evaluations and 10.80 s on the feature instances; comparable figures for the synthetic family). Tuning raises SG-dyn's standard-start success from 5/9 to 9/9 and, as the Reviewer anticipated, a tuned subgradient method can match or outperform DTAP — e.g. on Abalone, and on the nonconvex family tuned SG-const and tuned SG-dyn reach the tolerance in all 36 runs versus 24/36 for DTAP (Table 7). We have accordingly revised every comparative claim: the text now states that no pairwise comparison is significant after Holm adjustment, that the classification results reflect a size–prediction trade-off rather than superiority, and, in the Conclusion, that "tuned subgradient methods can match or outperform DTAP". The reported advantage of DTAP is narrowed to the fact that no modulus has to be estimated or tuned as an input of the iteration.

Response to Reviewer #2

We thank the Reviewer for the detailed, page-by-page reading. We have followed all comments; the paper is now substantially shorter and, we hope, considerably clearer.

Major issues

M1. Sections 1 (Introduction) and 2 (Related work) should be reorganized and merged into Section 1, limited to about 4 pages; the paper should be shortened (about 25 pages) by removing redundant content and long sentences.

Response. Done. The Related work section (and its Table 1) has been removed and its essential content condensed into the Introduction [Sec. 1], which now recalls the CQ method, self-adaptive and decomposition strategies, and the closest nonlinear antecedents (Li et al., Yen et al., Hu et al.) in a few short paragraphs and states three contributions instead of five. The Preliminaries were reduced to the facts actually used and no longer carry subsections; the feature-selection framework (former Section 5) is now a subsection of the experiments [Sec. 4.1]; the sparse variant DTAP-S, the runtime comparison, the robustness/ablation section, and the "Summary of findings" were removed; the Conclusion is a single paragraph. The paper is now 25 pages, down from 39.

M2. The writing should be substantially improved: many sentences are ambiguous or too long. Describe the problem, algorithms, etc. in short sentences; polish each sentence; remove "nonsense" descriptions.

Response. We rewrote the manuscript with this request in mind. Long sentences were split, ambiguous phrases were replaced or deleted, and descriptive passages that did not support a result were removed. The abstract, Introduction, Preliminaries, the opening of Section 3, and the entire experimental section were rewritten. The parenthetical names of lemmas, propositions, remarks, and corollaries were removed to shorten the text.

M3. The reference format is inconsistent, and important SFP references are omitted; e.g. the authors of [23] and [35] also have self-adaptive or decomposition work on the SFP.

Response. We have unified the reference format to one style (initials, italic journal names, volume, issue, pages, year, and DOI where available), and we now cite the self-adaptive and decomposition works of the authors of [23] (Z. Li, D. Han, W. Zhang) and [35] (Q. Yang) on the (multiple-sets) SFP: Zhang–Han–Li (*Inverse Problems*, 2009), Zhang–Han–Yuan (*Comput. Optim. Appl.*, 2012), Zhao–Yang (*Inverse Problems*, 2011), and Yang–Yang–Zhang (*J. Optim. Theory Appl.*, 2014). They are cited in the first paragraph of the Introduction, where self-adaptive steps and decomposition strategies for the SFP are recalled. References that were needed only for the removed material were deleted.

M4. Why are some numerical results presented in the Appendix?

Response. We agree, and the appendix has been removed. The per-dataset accuracy and AUC values are now in the main text [Table 3, Sec. 4.3], and the two baseline solvers are described in the main text [Sec. 4.2.1]. The threshold and runtime tables were removed; the thresholds of every instance are recorded in the supplementary reproducibility package.

Minor issues

Response. We agree with all twenty minor comments and have adopted every one of them without exception. For conciseness we list them, with the corresponding change and its location, in the table below; in each case the change is marked in colored text in the revised manuscript.

Reviewer's minor comment	**Change made (location in revised manuscript)**
m1 (p. 2, l. 18) — "reference point" is ambiguous.	Phrase removed; the López et al. rule is now cited among the self-adaptive methods in Sec. 1.
m2 (p. 2, l. 27) — "differing" vs. "different densities".	"Different"; the radiotherapy sentence was removed (Sec. 1).
m3 (p. 2, l. 35) — polish "The NSFP is not itself new".	Sentence removed; Li et al., Yen et al. and Hu et al. are now compared with DTAP directly (Sec. 1).
m4 (p. 2, l. 39) — "active index set" vs. "active set".	"Active set"; sentence removed; we use "approximate active set" (Sec. 3.2).
m5 (p. 2, ll. 47–49) — "We ask instead …" is unclear.	Removed; the class relation is restated in short sentences (Sec. 1).
m6 (p. 3, ll. 48–53) — sentences too long.	Paragraph removed; reduced to one sentence (Sec. 1).
m7 (p. 3) — five contributions too detailed.	Replaced by three short items C1–C3 (Sec. 1).
m8 (p. 5) — "Rule (3) has since been extended" grammar.	Sentence removed with the Related work section.
m9 (p. 5, l. 33) — "drives their proof is for free"; "nonconvexity sits in the sets".	Unclear passage removed with the Related work section.
m10 (p. 5, l. 35) — "plain gradient inequality".	Removed (it meant the convex gradient inequality $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$).
m11 (p. 5, l. 44) — "the closest predecessor".	Removed; the three antecedents are compared with DTAP directly (Sec. 1).
m12 (p. 5, l. 48) — "step" vs. "step size".	"Step size"; sentence removed with the Related work section.
m13 (p. 7, l. 31) — "dtap" should be "DTAP".	Corrected throughout; the acronym is now always "DTAP".
m14 (p. 7, l. 39) — "Combettes 10" should be "Combettes [10]".	Corrected; the surviving citation reads "in the sense of Combettes [5]" (Sec. 3.7).
m15 (p. 9) — subsections 3.1–3.2 unnecessary; 3.3 should move to the experiments.	Preliminaries de-subsectioned (Sec. 2); the feature-selection model is now Sec. 4.1.
m16 (p. 12) — $H := \mathbb{R}^n$ repeated.	Repetition removed from the problem statement (Sec. 3.1).
m17 (p. 13) — meaning of "standing assumption".	Subsection retitled "Assumptions" (Sec. 3.3); the name was dropped.
m18 (p. 18, l. 40) — AMS style for $\sum_{k=0}^{\infty} \|x^{k+1} - x^k\|^2 < \infty$.	Series displayed with limits above/below (Prop. 3.11); other displays checked.
m19 (p. 20, l. 36) — polish "so no Young splitting is needed".	Phrase removed from the proof of Corollary 3.13.
m20 (p. 20, l. 35) — expression $K \cap \bigcap_{i \in I^{\mathrm{cvx}}}$ is not good.	Rewritten as $\Omega = \{x \in K : F_i(x) \leq b_i \ \forall i \in I^{\mathrm{cvx}}\}$ (Cor. 3.13), and as $\bigcap_{i \in I^{\mathrm{cvx}}} \{x \in K : F_i(x) \leq b_i\}$ in the proof of Thm. 3.12.

General. Please consider the above comments and make the paper concise.

Response. We have followed every comment and shortened the paper from 39 to 25 pages (see M1 and M2).

Additional changes made by the authors

While revising, we also made the following changes, not requested by the Reviewers, which we list for completeness and transparency.

1. **Convexity of the fractional criterion.** The submitted version called $F_3(w) = w^\top Q_R w / \rho^\top w$ pseudoconvex. We now prove it is *convex* on an open neighbourhood of the simplex when $Q_R \succeq 0$ and $\min_k \rho_k > 0$ (Proposition 4.1), so the feature-selection instance is an all-convex NSFP. The assignment F_3 to the pseudoconvex track used in the experiments remains admissible (convex $\Rightarrow$ pseudoconvex), and the all-convex assignment is reported as a comparator.

2. **Domain of the component functions.** The functions are now defined on an open neighbourhood U of K, and (A2) requires continuous differentiability on U; this covers fractional functions with denominators positive on K. A short paragraph after Assumption 1 explains this and why Ω is convex.

3. **Scope of the analysis.** Besides Remarks 1 and 2 (see R1.1, R1.2), Remark 5 now states precisely that the $O(N^{-1/3})$ bound concerns only the *best* convex-track violation — not a last-iterate bound, not a rate for the full residual or the distance to feasibility.

4. **Experimental protocol.** Section 4.2.2 now states the protocol exactly: min–max scaling is fitted once on the training pool, the voted masks are computed once and reused, the SVM probability calibration uses a fixed seed, and the evaluation is exploratory rather than nested. The Nogueira stability index (with the correct $N_f/(N_f-1)$ variance correction) and the statistical procedure (exact signed-rank test, Holm adjustment, Friedman test) are given explicitly.

5. **Recomputed values.** The stability values, the SVM AUC values, and the p-values were recomputed and now include raw and Holm-adjusted p-values; columns for SG-const and SG-dyn were added. Accuracies and all 5-NN results are unchanged.

6. **Revised claims.** Statements of superiority (accuracy, speed, robustness) were removed or qualified: no pairwise comparison is significant after Holm adjustment, the results show a size–prediction trade-off, and tuned subgradient methods can match or outperform DTAP.

7. **Presentation of the algorithm.** The stopping test of Algorithm 1 is now stated at the start of Step 1 instead of in a separate Step 3; the iterates and the returned point are unchanged. Since all convergence statements are in $\mathbb{R}^n$, "strong convergence" is now written "convergence".

8. **Denominator safeguard.** Section 4.1.3 now reports that the smallest Fisher score on all instances is 4.77×10^{-5}, so the safeguard $\max\{\rho^\top w, 10^{-15}\}$ of the implementation is never active and the implemented F_3 is exactly (41).

9. **Declarations.** The declaration of generative-AI use and the data-availability statement were updated, and a reproducibility package now accompanies the paper.

We believe these revisions fully address the Reviewers' and Editor's comments and have improved the clarity, rigor, and honesty of the manuscript. We thank the Reviewers and the Editor again for their time.